

REPORT

Disease Prediction from Medical Data using Machine Learning

Project Overview

This project is a Machine Learning-based system that predicts the possibility of heart disease using structured medical data. Patient-related features such as age, sex, chest pain type, blood pressure, cholesterol level, heart rate and other medical attributes are used for prediction.

Different classification algorithms are trained and compared to identify the best-performing model for the selected dataset.

Objective

The main objective of this project is to develop a machine learning classification system that can analyze medical data and predict the target class associated with heart disease.

The project also aims to compare different machine learning algorithms based on their performance.

Key Features

- Medical dataset analysis
- Data validation and missing-value checking
- Data visualization
- Feature and target separation
- Train-test data splitting
- Multiple ML model implementation
- Accuracy comparison
- Confusion matrix evaluation
- Classification report
- Sample patient prediction

Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- XGBoost

Dataset

The dataset contains medical attributes such as:

- Age
- Sex
- Chest pain type
- Resting blood pressure
- Cholesterol
- Fasting blood sugar
- Resting ECG
- Maximum heart rate
- Exercise-induced angina
- Oldpeak
- Slope
- Number of major vessels
- Thal
- Target

The **target** column represents the class that the machine learning model tries to predict.

The heart.csv used in this academic demonstration is synthetic/educational data. It should not be considered clinical data.

Project Workflow

Medical Dataset



Data Loading



Data Checking



Data Visualization



Feature & Target Separation



Train-Test Split



Model Training



Prediction



Model Comparison



Confusion Matrix



Best Model Selection

Machine Learning Algorithms

1. Logistic Regression

Logistic Regression is a classification algorithm used to predict a categorical outcome.

In this project, it is used to classify the target into two classes.

2. Support Vector Machine (SVM)

SVM is a supervised machine learning algorithm that finds a decision boundary to separate different classes of data.

3. Random Forest

Random Forest uses multiple decision trees and combines their results to produce a final classification.

4. XGBoost

XGBoost is a gradient boosting algorithm that builds decision trees sequentially to improve prediction performance.

Implementation

Step 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

These libraries are used for data processing, numerical operations and visualization.

Step 2: Load Dataset

```
df = pd.read_csv("heart.csv")
```

The CSV file is loaded into a Pandas DataFrame.

Step 3: Check Dataset

```
print(df.shape)
print(df.columns)
print(df.isnull().sum())
```

This checks the number of rows and columns, column names and missing values.

Step 4: Visualization

```
sns.countplot(x='target', data=df)
plt.title("Heart Disease Distribution")
plt.show()
```

The graph shows the distribution of the target classes.

Step 5: Feature and Target Separation

```
X = df.drop("target", axis=1)
```

```
y = df["target"]
```

X contains the input features, while y contains the target value.

Step 6: Train-Test Split

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

The dataset is divided into training data and testing data.

Step 7: Train Models

Four classification models are trained:

- Logistic Regression
- SVM
- Random Forest
- XGBoost

Step 8: Prediction

The trained models predict the target class for the test data.

Step 9: Evaluation

The models are evaluated using accuracy, classification report and confusion matrix.

Model Performance

The four models are compared based on their test accuracy.

Model	Accuracy
Logistic Regression	Check Colab Output
SVM	Check Colab Output
Random Forest	Check Colab Output

XGBoost

Check Colab Output

The model with the highest accuracy is considered the best-performing model **for this experiment and dataset split**.

Evaluation Metrics

Accuracy

Accuracy represents the proportion of correctly classified samples.

Classification Report

The classification report provides:

- Precision
- Recall
- F1-score
- Support

Confusion Matrix

The confusion matrix shows how many samples were correctly and incorrectly classified into each class.

Sample Prediction

A sample patient record can be provided to the trained model.

```
prediction = rf_model.predict(new_patient)
```

```
if prediction[0] == 1:
```

```
    print("Prediction: Risk Detected")
```

```
else:
```

```
    print("Prediction: No Risk Detected")
```

The model returns the predicted class based on the given input features.

Advantages

- Simple and easy-to-understand implementation
- Uses multiple classification algorithms
- Allows model performance comparison
- Provides graphical evaluation

- Can be extended into a web application

Limitations

- The current dataset is synthetic and educational.
- Model performance depends on the dataset and train-test split.
- Accuracy alone is not sufficient for evaluating a real medical system.
- The system is not clinically validated.

Future Enhancements

- Use a validated real-world medical dataset
- Apply feature scaling and advanced preprocessing
- Perform hyperparameter tuning
- Use cross-validation
- Add ROC-AUC analysis
- Develop a web-based user interface
- Deploy the educational prototype online

Conclusion

The **Disease Prediction from Medical Data using Machine Learning** project demonstrates how classification algorithms can be applied to structured medical data.

Logistic Regression, SVM, Random Forest and XGBoost are implemented and compared. The models are evaluated using accuracy, classification reports and confusion matrices.

The project provides a basic understanding of the complete Machine Learning workflow, from dataset preparation to model evaluation and prediction.